



Graphs in Machine Learning

Spectral Clustering: RatioCut & NCut

Relaxation & Approximation Algorithms

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Partially based on material by: Ulrike von Luxburg,
Gary Miller, Doyle & Schnell, Daniel Spielman



Spectral Clustering: Relaxing Balanced Cuts

Relaxation for (simple) balanced cuts for 2 sets

$$\min_{A,B} \text{cut}(A, B) \quad \text{s.t.} \quad |A| = |B|$$

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$$\text{cut}(A, B) = \sum_{i \in A, j \in B} w_{i,j} = \frac{1}{4} \sum_{i,j} w_{i,j} (f_i - f_j)^2 = \frac{1}{2} \mathbf{f}^T \mathbf{L} \mathbf{f}$$

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What is the relationship with the **smoothness** of a graph function?

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objective function of spectral clustering

$$\min_{\mathbf{f}} \mathbf{f}^T \mathbf{L} \mathbf{f} \quad \text{s.t.} \quad f_i = \pm 1, \quad \mathbf{f} \perp \mathbf{1}_N, \quad \|\mathbf{f}\| = \sqrt{N}$$

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$$\cancel{f_i = \pm 1} \rightarrow f_i \in \mathbb{R}$$

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Rayleigh-Ritz theorem

If $\lambda_1 \leq \dots \leq \lambda_N$ are the eigenvalues of real symmetric $\mathbf{L}$ then

$$\lambda_1 = \min_{\mathbf{x} \neq 0} \frac{\mathbf{x}^T \mathbf{L} \mathbf{x}}{\mathbf{x}^T \mathbf{x}} = \min_{\mathbf{x}^T \mathbf{x} = 1} \mathbf{x}^T \mathbf{L} \mathbf{x}$$
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How can we use it?

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Generalized Rayleigh-Ritz theorem (Courant-Fischer-Weyl)

If $\lambda_1 \leq \dots \leq \lambda_N$ are the eigenvalues of real symmetric $\mathbf{L}$ and $\mathbf{v}_1, \dots, \mathbf{v}_N$ the corresponding orthogonal eigenvectors, then for $k = 1 : N - 1$

$$\lambda_{k+1} = \min_{\mathbf{x} \neq 0, \mathbf{x} \perp \mathbf{v}_1, \dots, \mathbf{v}_k} \frac{\mathbf{x}^T \mathbf{L} \mathbf{x}}{\mathbf{x}^T \mathbf{x}} = \min_{\mathbf{x}^T \mathbf{x} = 1, \mathbf{x} \perp \mathbf{v}_1, \dots, \mathbf{v}_k} \mathbf{x}^T \mathbf{L} \mathbf{x}$$

$$\lambda_{N-k} = \max_{\mathbf{x} \neq 0, \mathbf{x} \perp \mathbf{v}_N, \dots, \mathbf{v}_{N-k+1}} \frac{\mathbf{x}^T \mathbf{L} \mathbf{x}}{\mathbf{x}^T \mathbf{x}} = \max_{\mathbf{x}^T \mathbf{x} = 1, \mathbf{x} \perp \mathbf{v}_N, \dots, \mathbf{v}_{N-k+1}} \mathbf{x}^T \mathbf{L} \mathbf{x}$$

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objective function of spectral clustering (same - it's magic!)

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Spectral Clustering: Approximating NCut

Normalized Cut

$$\text{NCut}(A, B) = \sum_{i \in A, j \in B} w_{ij} \left(\frac{1}{\text{vol}(A)} + \frac{1}{\text{vol}(B)} \right)$$

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objective function of spectral clustering (NCut)

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$$\min_{\mathbf{w}} \mathbf{w}^T \mathbf{D}^{-1/2} \mathbf{L} \mathbf{D}^{-1/2} \mathbf{w} \quad \text{s.t.} \quad w_i \in \mathbb{R}, \quad \mathbf{w} \perp \mathbf{D}^{1/2} \mathbf{1}_N, \quad \|\mathbf{w}\|^2 = \text{vol}(\mathcal{V})$$

Spectral Clustering: Approximating NCut

objective function of spectral clustering (NCut)

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Spectral Clustering: Approximating NCut

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Spectral Clustering: Approximating NCut

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Spectral Clustering: Approximating NCut

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tl;dr: Get the second eigenvector of $\mathbf{L}/\mathbf{L}_{\text{rw}}$ for RatioCut/NCut.

Spectral Clustering: Approximating NCut

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tl;dr: Get the second eigenvector of $\mathbf{L}/\mathbf{L}_{\text{rw}}$ for RatioCut/NCut.

demo: <https://dominikschmidt.xyz/spectral-clustering-exp/>

Spectral Clustering: Approximation

These are all approximations. How bad can they be?

Spectral Clustering: Approximation

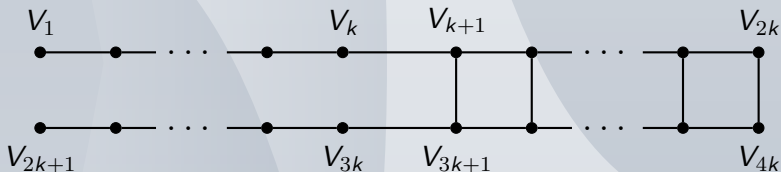
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Pretty bad.

Spectral Clustering: Approximation

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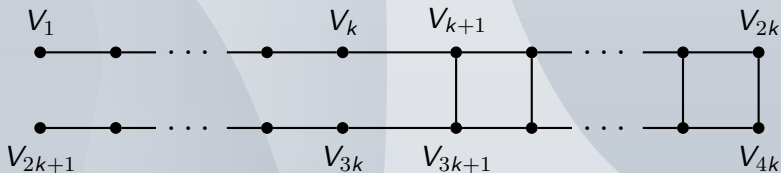
Pretty bad. Example: cockroach graphs



Spectral Clustering: Approximation

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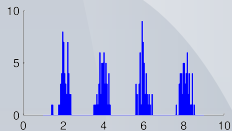
Pretty bad. Example: cockroach graphs



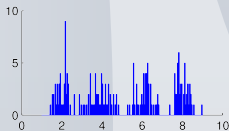
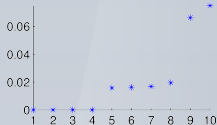
No efficient approximation exist. Other relaxations possible.

<https://www.cs.cmu.edu/~glmiller/Publications/Papers/GuMi95.pdf>

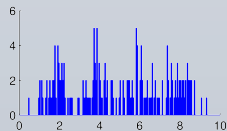
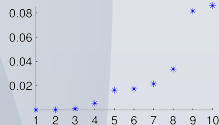
Spectral Clustering: 1D Example



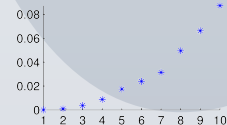
Eigenvalues



Eigenvalues

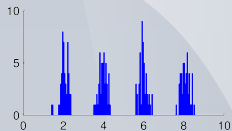


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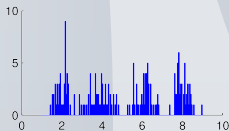
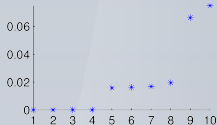


Source: von Luxburg (2007)

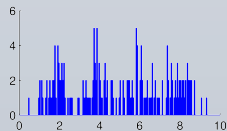
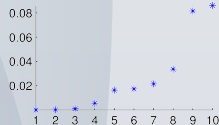
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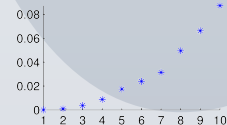
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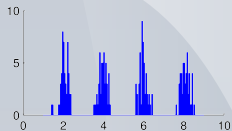


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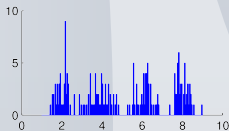
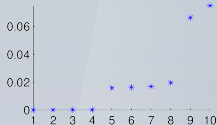


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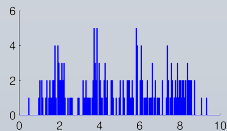
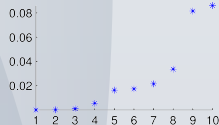
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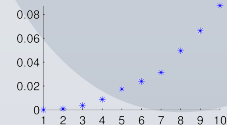
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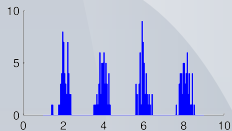


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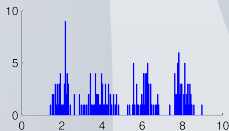
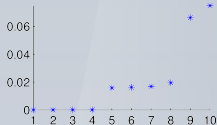


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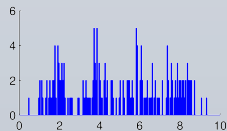
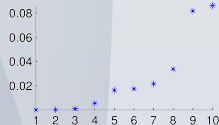
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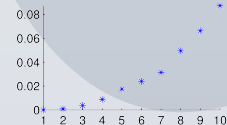
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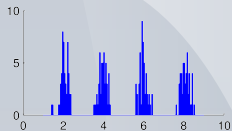
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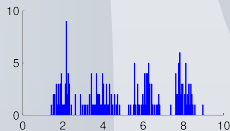
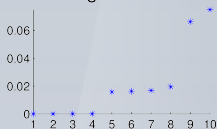
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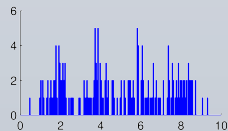
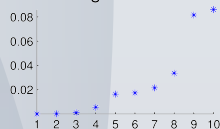
Elbow rule/EigenGap heuristic for number of clusters



Eigenvalues



Eigenvalues



Eigenvalues



Source: von Luxburg (2007)

Michal Valko

`michal.valko@inria.fr`

Isara Labs



<https://misovalko.github.io/mva-ml-graphs.html>